JADE COLLINS

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EDUCATION:

B.S Computer Science & B.A Biology *Fall 2024 - expected Spring 2027*
New Mexico State University *GPA: 4.0*
Associate of General Studies *Fall 2022 - Spring 2024*
Received Dec 2025
Colorado State University *GPA: 3.96*

LANGUAGES:

English - Advanced; Spanish - Beginner

WORK EXPERIENCE:

Bailey Lab Undergraduate *August 2024 - present*
New Mexico State University, Las Cruces, NM

- Conducted bioinformatic and phylogenetic research on the evolutionary history of mimosine production in plants using DIAMOND and OrthoFinder softwares.
- Managed genomic data on a Linux server while researching the evolution of novel chemistry.
- Presented my research at 5 conferences and symposiums in both poster and talk form: MANRRS National Conference, NMSU Biosymposium, NMSU URCAS, NMSU AMP, and SFA^{2}F.
- Awarded Best Undergraduate Oral Presentation at NMSU Biosymposium.

Agricultural Communications Specialist *May 2026 - present*
New Mexico State University, Las Cruces, NM

- Oversaw content strategy for 3 social media identities across 7 platforms spanning university-wide to multi-state audiences, covering topics in agriculture, education, and science.
- Adapted 30+ pieces of technical scientific content into accessible social media posts tailored for both scientific and general audiences, contributing to increased account interactions and website traffic.
- Captured images, videos, and interviews offsite and on set.
- Led and created a promotional campaign for New Mexico's AgrAbility program, working with the program coordinator on her vision.
- Tested and documented 2 in-development educational games, providing structured feedback that informed the addition of a help feature and inclusive character design.
- Taught 20+ middle school and high school children graphic design in a classroom space.
- Mentored four 4-H members on media capture, management, and editing, producing 8 posts over 3 days for a promotional campaign for the New Mexico 4-H State Conference.

CodePath Tech Fellow *Aug 2025 - Dec 2025*
CodePath, Las Cruces, NM

- Mentored and led 70+ first-year computer science students at New Mexico State University in intro to computer science, helping them build foundational technical skills and confidence.
- Facilitated weekly office hours and lab sessions, guiding students through testing, troubleshooting, and technical problem solving while promoting teamwork and critical thinking.
- Graded assignments with precision and consideration.

YWiC Student Aide *Aug 2024 - Aug 2025*

CS Dept., New Mexico State University, Las Cruces, NM

- Developed and taught CS curriculum covering computer vision, AI, and robotics for programs serving 100+ elementary, middle, and high school students across Southern NM.
- Planned and managed outreach events 5+ times a semester for underrepresented individuals in computer science while fostering a CS community at NMSU.

Lauder Lab Undergraduate *June 2024 - Aug 2024*

Harvard University, Cambridge, MA

- Developed R program to run boundary layer analysis.
- Prepared and analyzed shark skin samples: used GelSight to image surface of skin samples; used MountainsMap software to analyze GelSight images; set up and ran experiments using a flow tank and high-speed camera software; used PIV (particle image velocimetry) and analysis software DaVis to analyze data collected.
- Created STL models for 3D printing.
- Synthesized data and current results into a poster and presented it at SICB 2025.

Student Org Specialist *May 2023 - May 2024*

SLiCE at Colorado State University, Fort Collins, CO

- Manage annual Registered Student Organizations (RSO) registration process.
- Assist RSO members, officers, & advisors in navigating/utilizing RamLink – CSU's online involvement hub.
- Guide members, officers, & advisors through varying university policies, procedures, & opportunities.
- Coordinate, facilitate, & create marketing material for educational trainings, workshops, retreats, & annual award event.
- Maintain inventory & upkeep within RSO resource center – the Student Organization Center.
- Curate and distribute monthly student organization newsletter to 18,000+ students and faculty.

SIGNAL Lab Undergraduate *Feb 2023 - May 2024*

Colorado State University, Fort Collins, CO

- Organized and labeled 1000s of images for 2+ research papers, improving machine learning models by correcting bounding boxes.
- Created 1000+ human action video image annotations to establish datasets for machine learning on 2+ projects.
- Participated in 3 data-collection studies for machine learning and AI, increasing the quality of the projects.

OURA Lab Undergraduate *Jan 2023 - Apr 2023*

Colorado State University, Fort Collins, CO

- Worked in a team and delegated tasks to complete multiple experiments.
- Practiced lab safety and built strong lab technique basics.
- Designed, ran, and presented own experiment on bacteria growth.
- Compiled literature review and presented findings and own hypothesis on harmful algal blooms at CSU

MURALS 2023 and 2024.

Dining Hall Team Member *Aug 2022 - Apr 2023*
Colorado State University, Fort Collins, CO

- Prepare food, cook food, serve customers, maintain health standards of a food workplace, work the register, restock food in a marketplace, and barista responsibilities.
- Uphold customer service standards.
- Acting supervisor when none present.

Assistant Teacher & Camp Leader for YWiC *2022, 2023, 2025*
CS Dept., New Mexico State University, Las Cruces, NM

- Introduced computer science concepts like ML, AI, VR, app dev, web dev, cyber security, and binary.
- Managed and taught 20-30 students (middle school and high school) for 2 week-long summer camps each.

VOLUNTEER WORK:

Tortugas Reptile Rescue July 2026 - *present*
Las Cruces, NM

- Rescue, rehabilitate, and rehome reptiles in Las Cruces and the surrounding areas.
- Clean, setup, and monitor animal enclosures.
- Feed and provide typical and medical care for animals.
- Assist with educational events and workshops with ambassador animals.

El Paso Zoo and Botanical Gardens June 2025 - *present*
El Paso, TX

- Animal Handler with the Animal Ambassador Education dept.
- Assist with special event preparation, running, and take down
- Prepared animal diets such as zebra, kudu, giraffe, pronghorn, rhino, and hogs
- Responsible for ~30 elementary school children during Zoo Camp
- Greeted guests and presented animal facts
- Cleaned incoming possum exhibit

New Mexico Game and Fish Department *Dec 2024 - May 2025*
New Mexico

- Standby volunteer for various activities
- Set up turkey camera traps in Gila National Forest
- Assist with New Mexico Nightjar annual survey

CSUnity *Apr 2024*
Fort Collins, CO

- 4 hours of volunteer work helping the community
- Individuals and businesses in the community apply to get help for this day
- Volunteer teams are assigned to a community project
- My team helped the Fort Collins Trolley clean the trolleys and depot

Project Homeless Connect *Apr 2024*
Fort Collins, CO

- 2 hours of volunteer at a community center

- For one day, businesses around the community volunteer to provide free services to homeless individuals
- I helped guide those who needed help and take down at the end of the day

Technology Class for Older Adults *Sept 2023 - Apr 2024*
Fort Collins, CO
- 2 hour class every other week focusing on different technology topics
- Older adults sign up for these classes and I volunteer to help teach them one-on-one at the Fort Collins Senior Citizen Center

Ag Day BBQ *Sept 2023*
Colorado State University, Fort Collins, CO
- Helped take down and put away chairs and tables for Ag Day BBQ at Colorado State University

Chemistry Club Halloween Show *Oct 2022*
Colorado State University, Fort Collins, CO
- Manned a booth at the Halloween show, teaching young kids and their families (1000+) in the community about science through a fun and simple project
- Helped set up and take down the tables and decorations

PROJECTS:

Pawspect | *2026*
Technologies: Python, Tensorflow, Docker, Google Colab, GitHub
- Built an AI-powered cat \& dog breed identification web app as part of AI4ALL Ignite, achieving 95% and 97% top-3 CNN model accuracy for cat and dog models, respectively.
- Led a team of 8 members as project lead, coordinating task assignments, sprint planning, and presentations across development, evaluation, and documentation phases.
- Preprocessed and combined datasets of 25,000+ images, developed and optimized a CNN model reducing error by 75%, and containerized the application using Docker for consistent deployment.

Muscle Meow Website | *2025*
Technologies: Python, SQL, Streamlit, Google Cloud Console, GitHub
- Built and deployed a full-stack workout app website hosted on Google Cloud Platform with a CD pipeline via GitHub Actions as part of a Google Tech Exchange team project.
- Planned, prototyped, and conducted user testing using Figma, identifying 3 usability issues that drove navigation and layout restructuring.
- Designed and implemented UI and features for 3 pages including an automatic workout logger, social feed, and activity summary using Python and Streamlit.

SongSync | *2024*
Technologies: Java, GitHub, VSCode
- Java program developed with Agile methodology that can tell a user what songs they would like based on how they and other users rated the songs. Songs can be grouped by similarity based on ratings.
- Read database with song names and database with each user's ratings of each of the songs and used K-means clustering to categorize them.
- Used test-driven methodology. Code tests created in Java, performed with Gradle.

PROGRAMS:

AI4ALL Ignite | *Summer, Fall 2026* | *Remote*
A two-semester AI project-based course with technical instruction and career development lectures taught by industry professionals
Google Tech Exchange | *Spring 2025* | *Remote*
Virtual classes in Software Engineering with Generative AI, Algorithms and System Design, and Technical Interview Prep taught by Google Software Engineers
Computer Vision Workshop | *Winter 2025* | *SICB (Society for Integrative and Comparative Biology) Conference*
Computer vision for pollinator biology workshop
Bioinformatics Workshop | *Summer 2024* | *Harvard University*
2 day bioinformatics workshop in R

TECHNICAL SKILLS:

Computer Science Languages:

Python, Java, HTML, CSS, JavaScript, R, C, SQL

Developer Tools:
GitHub, VSCode, Google Cloud Console, MySQL Workbench, Streamlit, Docker, TensorFlow, Raspberry Pi 4, Figma

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LABORATORY SKILLS:

Techniques:
Gel electrophoresis; Chemical synthesis, distillation, extraction, vacuum filtration, and recrystallization; Thin Layer Chromatography (TLC); Bacteria growth and transformation; Gene editing; Particle Image Velocimetry (PIV); Dissection

Tools:
DIAMOND, OrthoFinder, NCBI Blast, Spectrometer, GIS (Geographic Information System)

CONFERENCE PRESENTATIONS:

Collins, J. "Is that Blood in the water? The damaging effects of red tides" *Oral & Poster Presentation, Colorado State University Multicultural Undergraduate Research, Art, and Leadership Symposium (MURALS), Fort Collins, CO, 2023.*
Collins, J. "Is that Blood in the water? The damaging effects of red tides" *Oral & Poster Presentation, Colorado State University Multicultural Undergraduate Research, Art, and Leadership Symposium (MURALS), Fort Collins, CO, 2024.*
Collins, J., Law, D., Gabler-Smith, M. K., Lauder, G. V. "The function of shark skin denticles studied using biomimetic models" *Poster Presentation, The Society of Integrative and Comparative Biology (SICB), Atlanta, GA, 2025.*
Collins, J. & Bailey, C. D. "Evolutionary history of mimosine production in agriculturally significant legumes" *Poster Presentation, Minorities in Agriculture, Natural Resources, and Related Sciences (MANRRS) National Conference, Memphis, TN, 2025.*

Collins, J. & Bailey, C. D. "Evolutionary history of mimosine production in agriculturally significant legumes" *Poster Presentation, New Mexico State University Biosymposium, Las Cruces, NM, 2025.*
Collins, J. & Bailey, C. D. "Evolutionary history of mimosine production in agriculturally significant legumes" *Oral Presentation, New Mexico State University Undergraduate Research & Creative Arts Symposium (URCAS), Las Cruces, NM, 2025.*
Collins, J. & Bailey, C. D. "Evolutionary history of mimosine production in agriculturally significant legumes" *Poster Presentation, New Mexico State University Alliance for Minority Participation (AMP), Las Cruces, NM, 2025.*
Collins, J. & Bailey, C. D. "Evolutionary history of mimosine production in agriculturally significant legumes" *Oral Presentation, New Mexico State University Biosymposium, Las Cruces, NM, 2026.* **(Best Undergraduate Oral Presentation)**
Collins, J. & Bailey, C. D. "Evolutionary history of mimosine production in agriculturally significant legumes" *Oral Presentation, New Mexico State University Undergraduate Research & Creative Arts Symposium (URCAS), Las Cruces, NM, 2026.*
Collins, J. & Bailey, C. D. "Evolutionary history of mimosine production in agriculturally significant legumes" *Poster Presentation, SEQUENCE TO FUNCTION: Applications & Analysis for the Future (SFA2F), Santa Fe, NM, 2026.*
Collins, J. & Bailey, C. D. "Evolutionary history of mimosine production in agriculturally significant legumes" *Poster Presentation, New Mexico State University Alliance for Minority Participation (AMP), Las Cruces, NM, 2026.*
Collins, J. & Bailey, C. D. "Evolutionary history of mimosine production in agriculturally significant legumes" *Poster Presentation, Emerging Researchers National (ERN) Conference in STEM, Atlanta, GA, 2027. (Accepted)*

CURRENT MEMBERSHIPS:

National Technical Honors Society
National Society of Leadership and Success
National Society of Collegiate Scholars
National Center for Women & Information Technology
The Society for Collegiate Leadership & Achievement
ColorStack
WAFWA Student Engagement Program
TRIO STEM-H - NMSU

AWARDS & HONORS:

2022, 2025, 2026 Hispanic Scholarship Fund Scholar

2023, 2024 Hispanic Scholarship Fund Finalist

2022-2023, 2023-2024 College of Natural Sciences Dean's List

2022-2023, 2023-2024 MURALS Participant

2023-2024 4.0 GPA Academic Scholar

2024-2025, 2025-2026, 2026-2027 College of Arts and Sciences Dean's List

2025, 2026 URCAS Participant

2025 Discovery Scholars Program Scholar

2025 Biosymposium Participant

2025 Google Tech Exchange Participant

2026 NMSU Biosymposium Best Undergraduate Oral Presentation

2026 AI4ALL Ignite Fellow

SCHOLARSHIPS:

2022-2023, 2023-2024 National Merit Scholarship Awardee

2022-2023, 2023-2024 Colorado State University Honors Scholarship

2022-2023, 2023-2024 Colorado Opportunity Scholarship

2023-2024 ARCS Fund Scholarship - Natural Sciences

2023-2024 Lamb/Donar Undergrad Scholarship

2023-2024 Colorado State University Honors Outstanding Scholar

2023-2024 Billie&Dean Crouse/Acacia Scholarship

2024-2025 AI Education Tuition

2024-2025 Federal Pell Grant

2024-2025, 2025-2026 Hadley Honors Transfer

2024-2025, 2025-2026 NSF/Crest-Center

2024-2025, 2025-2026, 2026-2027 New Mexico Opportunity Scholarship

2024-2025 Richard H Stark Scholarship

2024-2025 S-STEM

2025-2026 NMSU Scholarship Fund

2025-2026 Brandy K. Streit Memorial Scholarship

2026-2027 Eugene Staffeldt Memorial Scholarship

2026-2027 Jackie & Adrian Edwards Scholarship